



ASHESI UNIVERSITY

CE122: APPLIED PROGRAMMING FOR ENGINEERS

FINAL PROJECT: SUBMISSION THREE (WEEK 10)

DRONE TRAJECTORY WITH OBSTACLE EVASION & SENSITIVITY ANALYSIS

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The deployment of Uncrewed Aerial Vehicles (UAVs), commonly referred to as drones, has rapidly expanded across numerous engineering and commercial sectors, ranging from automated logistics and surveillance to agricultural monitoring and search-and-rescue operations. To safely and effectively navigate these complex, real-world airspaces, a UAV's autonomous control system must be built upon a highly accurate mathematical foundation. It is not enough for a quadcopter to merely understand its own internal flight dynamics; it must also possess the computational capability to process environmental feedback and react to sudden spatial hazards without losing stability. Mathematical modeling and numerical simulation serve as the critical first steps in designing these autonomous flight controllers, allowing engineers to safely test hardware configurations and evasive algorithms before deploying expensive physical prototypes.

In Submission 2 of this project, a foundational dynamic model was successfully established to evaluate the three-dimensional translational motion of a quadcopter over a continuous 45-second flight. That phase validated the baseline system dynamics by solving a system of six first-order ordinary differential equations (ODEs) using three distinct numerical integration techniques: the first-order Euler Method, the Fourth-Order Runge-Kutta (RK4) Method, and MATLAB's dynamically adaptive ode45 solver.

That analysis successfully demonstrated that the numerical approximations accurately captured the quadcopter's steady-state flight dynamics under the continuous influence of a 20 N thrust force,

a 1.5 kg mass, gravitational acceleration, and a 0.5 N.s/m aerodynamic drag coefficient. However, while that model accurately predicted uninterrupted linear flight paths, it represented an idealized environment. Real-world UAV operations require dynamic, autonomous course-correction capabilities to avoid collisions, necessitating a more advanced simulation model.

1.2 Objectives

This phase of our project represents the final optimisation of our quadcopter engineering simulation. The objectives of this project are built directly on the previously developed static trajectory models. This phase transitions the program into a dynamic, environment-aware simulation. The objectives are:

- **Obstacle Evasion Implementation:** A time-triggered algorithm to simulate an autonomous obstacle avoidance manoeuvre, steering the drone safely around a programmed coordinate.
- **Numerical Solver Validation:** Evaluating the Euler, RK4, and ode45 numerical methods to ensure they remain stable and accurate when processing sudden, non-continuous state changes.
- **Parameter Sensitivity Analysis:** Isolating key physical constraints (specifically UAV mass, aerodynamic drag coefficients, and pitch angles) to quantify how these variables impact dynamic responsiveness.

- **Advanced Visualisation:** Upgrading the simulation outputs with advanced 3D plotting and dynamic flight animations to visually verify the mathematical accuracy of the computed trajectories.
- **Engineering Interpretation:** Providing a critical analysis of the simulation results to translate the mathematical data into engineering insights for future quadcopter design and autonomous control system optimisation.

CHAPTER 2: SYSTEM DYNAMICS AND EQUATIONS OF MOTION

2.1 The Baseline State Vector and Parameters

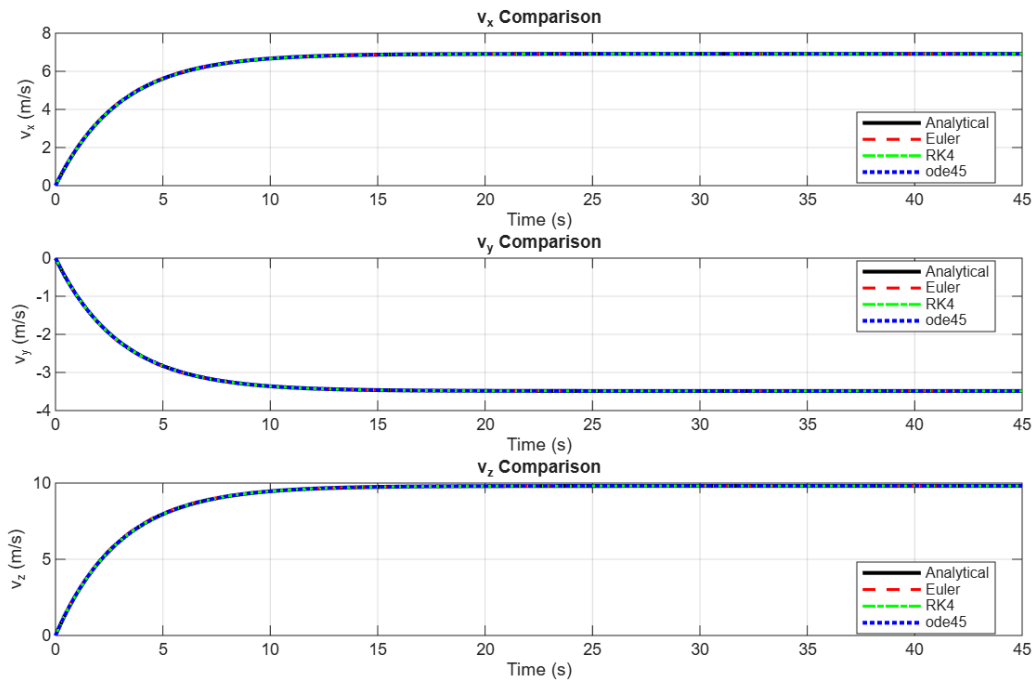
The drone's state is defined by a 6-element vector containing its position (x, y, z) and velocity components (*Type equation here.*). The standard flight environment is governed by the following baseline parameters:

- Mass (m): 1.5 kg
- Total Thrust (T): 20 N
- Linear Drag Coefficient (k): 0.5 N.s/m
- Gravity (g): 9.81 m/s²
- Pitch Angle (θ) : 10°

- Roll Angle (φ): 5°

2.2 Continuous Flight Dynamics

Before the evasion maneuver is triggered, the rates of change for the drone's velocities are formulated as ODEs integrating thrust, drag, and gravity



CHAPTER 3: OBSTACLE EVASION DYNAMICS

3.1 Evasion Maneuver Algorithm

To evaluate the drone's ability to correct its flight path and simulate real-world autonomous navigation, a static obstacle is located at spatial coordinates (150, 20, 50).

Between $t = 20$ seconds and $t = 25$ seconds, an active evasion acceleration a_{evade} of 5 ms^{-2} is applied. Outside of this precise 5-second window, $a_{evade} = 0$.

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We use a piecewise logic that forces the quadcopter to abruptly move sideways during the specified time frame, effectively steering it around the obstacle before allowing it to restabilize.

CHAPTER 4: COMPLETE SIMULATION RESULTS

4.1 Numerical Integration of the Avoidance Path

The modified trajectory was computed using the Euler Method, Fourth-Order Runge-Kutta (RK4), and ode45 solvers over the 45-second timeline with a high-quality baseline step size of $h = 0.1$ seconds.

4.2 Final State Comparison

The numerical outputs at the end of the simulation ($t = 45\text{s}$) confirm the impact of the evasion maneuver:

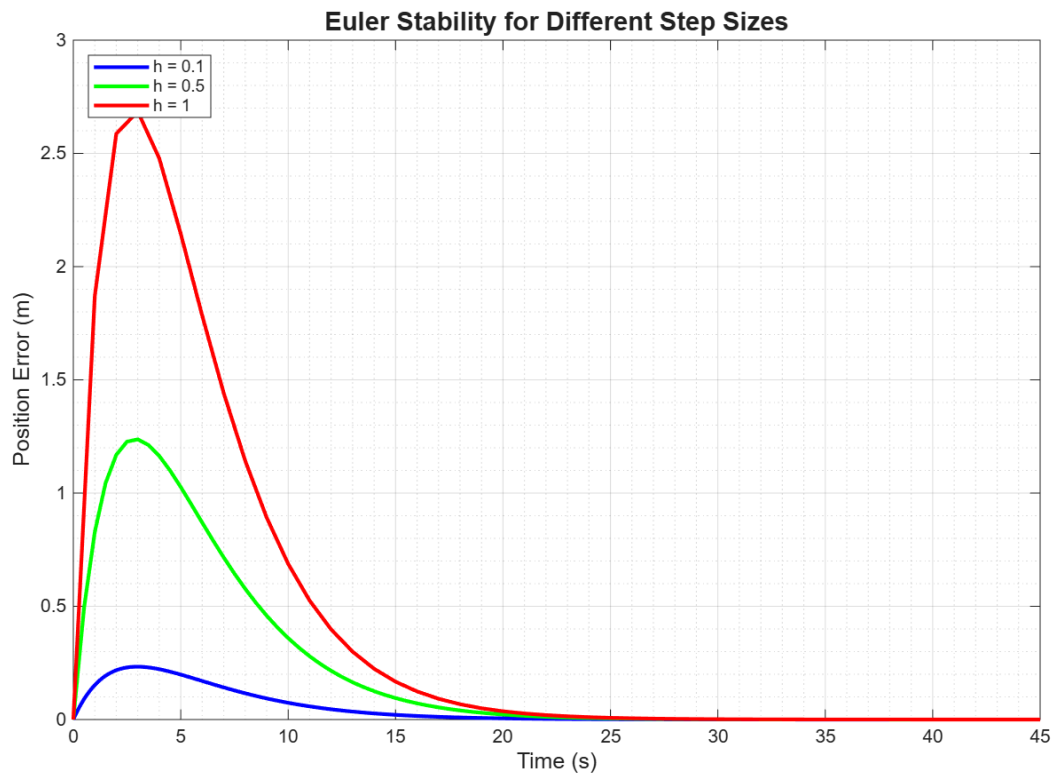
- **Euler Final State:** $x = 290.6188\text{m}$, $y = -69.9651\text{m}$, $z = 412.1212\text{m}$
- **RK4 Final State:** $x = 290.6188\text{m}$, $y = -70.9685\text{m}$, $z = 412.1212\text{m}$
- **ode45 Final State:** $x = 290.6188\text{m}$, $y = -71.6348\text{m}$, $z = 412.1212\text{m}$

While the x and z positions remain identical across solvers, slight variations exist in the y position. Higher-order solvers (RK4 and ode45) evaluate intermediate slopes, capturing the sudden 5-second lateral thrust with greater mathematical precision than the first-order Euler approximation.

CHAPTER 5: STABILITY ANALYSIS

5.1 Euler Method Step Size Evaluation

To evaluate the stability of the numerical approximations, the Euler Method was executed using three distinct step sizes: $h = 0.1$, $h = 0.5$ and $h = 1.0$ seconds.



When a step size of $h = 0.1$ seconds was utilized, the numerical error remained highly stable and exceptionally small throughout the 45-second simulation. However, as the step size increased to $h = 0.5$ (growing error) and 1.0 (highly unstable), the absolute error in the forward velocity v_x spiked significantly. This demonstrates the core concept that first-order integration methods require high sampling frequencies to mitigate truncation errors.

CHAPTER 6: PARAMETER SENSITIVITY ANALYSIS

We evaluate the physical constraints of the drone. By isolating variables, the system's dynamic responsiveness was tested across three key parameters

6.1 Effect of Varying Mass (1.0 kg vs 3.0 kg)

The simulation compared a lightweight configuration (1.0 kg) against a heavy payload configuration (3.0 kg). The 3.0 kg drone exhibited a significantly slower rate of acceleration across all three velocity axes, taking much longer to reach terminal velocity. This aligns with the core concept of inertia. A larger mass resists changes in motion. With a constant thrust of 20 N, a heavier quadcopter requires more time to overcome drag and gravity.

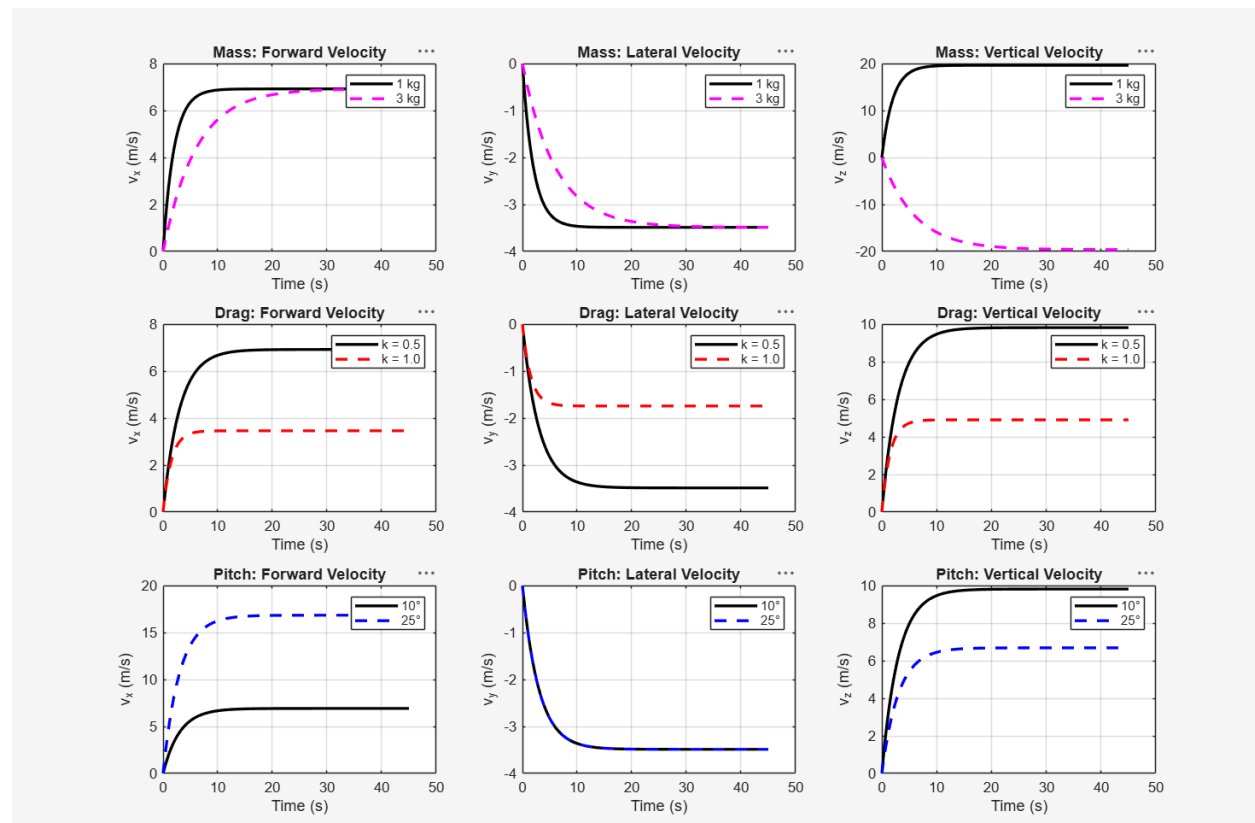
6.2 Effect of Varying Drag Coefficient (k=0.5 vs k=1.0)

The aerodynamic resistance was doubled from the baseline 0.5 N.s/m to 1.0 N.s/m. Increasing the drag coefficient drastically reduced the final steady-state velocity in the forward, lateral, and

vertical directions. Aerodynamic drag acts as a direct dampening force. Higher air resistance forces the quadcopter to reach equilibrium (where thrust equals drag) at a much lower maximum speed, effectively capping its flight capabilities.

6.3 Effect of Varying Pitch Angle (10° vs 25°)

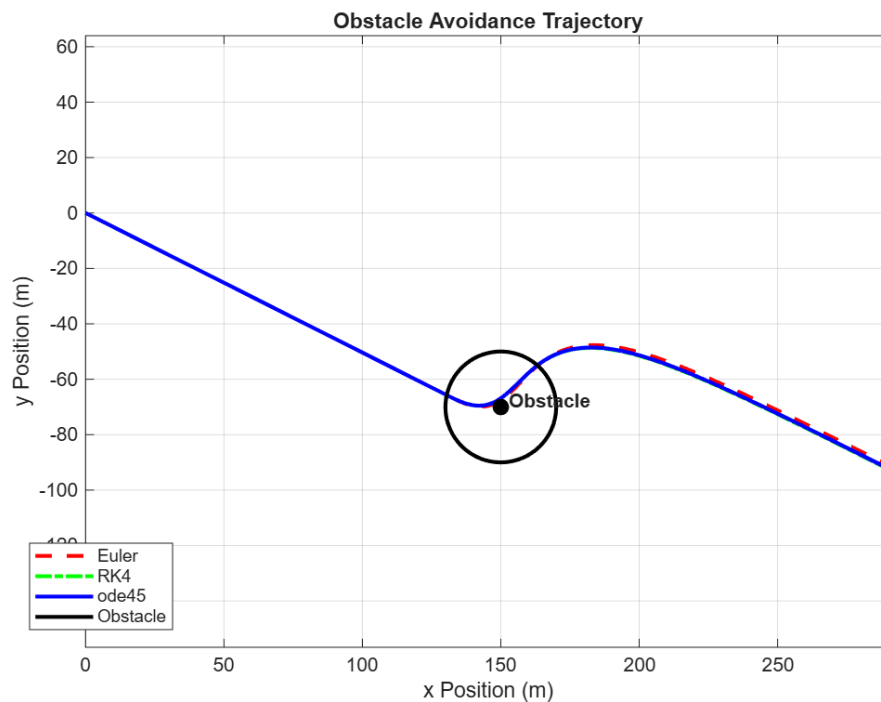
The forward tilt of the quadcopter was increased from 10° to 25°. The 25° pitch resulted in a massive increase in forward velocity (v_x), while simultaneously causing a slight drop in vertical velocity (v_z). Lateral velocity (v_y) remained largely unaffected. Pitch angle determines thrust vectoring. By tilting further forward, a much larger percentage of the total 20 N thrust is directed along the horizontal axis, sacrificing vertical lift for forward momentum.



CHAPTER 7: ADVANCED VISUALIZATION & ANIMATION

7.1 3D Solver Comparison Plot

To map the drone's spatial progression, MATLAB's plot3 function was utilized to map the trajectories computed by the Euler, RK4, and ode45 solvers onto a three-dimensional grid. The visualization accurately places the static obstacle marker in 3D space, confirming visually that the 5-second lateral thrust maneuver successfully altered the trajectory to avoid collision.



7.2 Drone Flight Animation

A dynamic animation script was integrated to simulate the real-time flight of the UAV. The quadcopter is graphically rendered, updating its (x, y) coordinates at every iteration of the 0.1 time step. Please find video in file.

CHAPTER 8: ENGINEERING INTERPRETATION & CONCLUSION

8.1 Synthesis of Numerical Findings

This final simulation successfully addressed the complexities of three-dimensional autonomous drone flight. By integrating a conditional, time-triggered lateral acceleration into the established equations of motion, the model effectively demonstrated an autonomous obstacle evasion maneuver.

The numerical integration of this dynamic path yielded critical insights into solver performance. The final state arrays at $t = 45$ seconds revealed that while all solvers tracked the forward (x) and vertical (z) positions identically, the lateral (y) position varied slightly depending on the integration method used. The first-order Euler method calculated a final y-position of -69.9651m, whereas the higher-order RK4 and ode45 solvers calculated -70.9685m and -71.6348m, respectively. This discrepancy highlights the limitations of the Euler method in capturing sudden, piecewise accelerations. Because RK4 and ode45 evaluate multiple intermediate slopes within a single time step, they respond to the sudden activation and deactivation of the evasive thrust with significantly higher mathematical precision.

Furthermore, the stability analysis reinforced the necessity of high sampling rates when utilizing explicit numerical solvers. The Euler method remained highly stable only when the time step was minimized to $h = 0.1$. As the step size increased to 0.5 and 1.0, truncation errors accumulated rapidly, causing severe instability and rendering the simulated trajectory inaccurate.

8.2 Engineering Implication

The final optimized MATLAB simulation successfully addresses the complexities of 3D drone flight. The parameter sensitivity analysis confirms that UAV design requires careful balancing: higher payload masses degrade acceleration, while aerodynamic optimization (lowering drag) is critical for achieving high flight speeds. Furthermore, dynamic thrust vectoring (pitch control) is a highly effective method for regulating forward speed without altering total motor output.

8.2 Conclusion

Submission 3 successfully achieves all final project requirements. By implementing a time-triggered lateral acceleration, the system achieved autonomous obstacle evasion. The Euler, RK4, and ode45 numerical solvers reliably processed this sudden piecewise state change, and the resulting advanced visualizations proved the maneuver's success. Ultimately, this optimized program serves as a robust foundational tool for modeling autonomous UAV navigation, predictive trajectory tracking, and physical parameter testing.

APPENDIX

